

PART I - FOUNDATION PLC PROJECTS

Project 2 - Automated Box Counting System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

Prepared By: Sharmin Rahman

1. Project Overview

The Automated Box Counting System was developed to introduce event-based counting using a PLC Count Up (CTU) instruction. The application represents a simple production-counting process in which boxes pass a sensor and the PLC keeps track of the number detected.

The system uses digital input I:1/0 as the box-detection signal and counter C5:0 to maintain the current count. The counter preset is set to 10. When the accumulated count reaches the preset value, the counter Done bit becomes true and energizes Green Lamp O:2/0 to indicate that the required batch quantity has been reached.

A separate Reset input I:1/1 executes a RES C5:0 instruction so the operator can clear the accumulated count and begin a new counting cycle.

This project extended my PLC foundation from time-based control into event-based control and introduced the relationship between a counter accumulator, preset value, Done status, and reset operation.

2. Engineering Objective

The objective of this project was to develop and verify a PLC program capable of counting individual box-detection events, maintaining the accumulated count in C5:0, indicating completion when 10 boxes had been counted, and resetting the counter and completion indication when requested.

The project also gave me practical experience using CTU, RES, counter status bits, and discrete output logic.

3. Functional Requirements

- The PLC shall monitor Box Sensor I:1/0.
- Counter C5:0 shall increment for each valid false-to-true transition of the counting rung.
- The counter preset shall be 10.
- Green Lamp O:2/0 shall remain OFF while the accumulated count is below the preset.
- When C5:0.ACC reaches 10, C5:0/DN shall become true.
- Green Lamp O:2/0 shall energize when C5:0/DN is true.
- Reset Button I:1/1 shall execute RES C5:0.
- Resetting the counter shall return the accumulated count to zero and clear the Done status.
- Clearing the Done status shall de-energize the Green Lamp.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the counting logic
Input Module	1762-IQ16	Provides discrete sensor and reset inputs
Output Module	1762-OB16	Provides the lamp output
Box Sensor	I:1/0	Provides box-detection events
Reset Button	I:1/1	Resets the counter
Box Counter	C5:0	Maintains the accumulated box count
Green Lamp	O:2/0	Indicates that the preset count has been reached
Programming Software	RSLogix Micro Starter Lite 8.30	Ladder logic development and monitoring

The Version 1 ladder logic does not control the conveyor itself. The movement of boxes past the sensor is treated as part of the surrounding process or simulation environment.

5. I/O Assignment

Physical I/O

Device	PLC Address	Type	Description
Box Sensor	I:1/0	Digital Input	Detects each box-counting event
Reset Button	I:1/1	Digital Input	Resets the box counter
Green Lamp	O:2/0	Digital Output	Indicates batch completion

Internal PLC Data

Function	Address	Type	Purpose
Box Counter	C5:0	CTU Counter	Stores the accumulated box count
Counter Preset	C5:0.PRE	Counter Data	Target quantity of 10
Counter Accumulator	C5:0.ACC	Counter Data	Current counted quantity
Counter Done	C5:0/DN	Status Bit	Becomes true when accumulated count reaches preset

6. Control Strategy

I used a three-function control strategy: counting, batch-complete indication, and reset.

Box Sensor I:1/0 drives Count Up counter C5:0. The Done status C5:0/DN controls Green Lamp O:2/0, while Reset Button I:1/1 executes RES C5:0.

Separating these functions made the program easy to follow. The counting rung handles event accumulation, the output rung handles completion indication, and the reset rung clears the counter independently.

An important characteristic of the CTU instruction is that it increments when the rung transitions from false to true. A sensor remaining continuously true across multiple PLC scans does not continuously increment the counter; a new false-to-true transition is required for another count.

7. Engineering Design Decisions

Design Decision	Reason
Used CTU C5:0	The application required event-based counting rather than timing
Set the preset to 10	The required batch quantity was 10 boxes
Used C5:0/DN for completion indication	The Done bit directly represents whether the preset has been reached
Controlled the Green Lamp separately	This kept counting independent from output indication
Used RES C5:0	A new batch requires accumulated count and counter status to be cleared
Kept Version 1 logic to three functional rungs	The objective was to understand CTU operation without unnecessary complexity

The Version 1 program intentionally does not stop a conveyor, log production data, display the current count on an HMI, or validate sensor quality. Those functions are better addressed in the industrial Version 2 design.

8. Ladder Logic Description

The Version 1 program contains three functional rungs. Important logic is shown in the approved clean single-line format and is supported by the original RSLogix implementation.

8.1 Rung 0000 - Count Boxes

```
|----[ XIC I:1/0 ]-----[ CTU C5:0 ]----|
      Box_Sensor              PRE: 10   Box_Counter
```

When Box Sensor I:1/0 produces a false-to-true transition, CTU C5:0 increments its accumulated value by one. The counter preset is 10, and the accumulator retains the current count until the counter is reset.

8.2 Rung 0001 - Batch Complete Indication

```
|----[ XIC C5:0/DN ]----- ( OTE O:2/0 )----|
      Box_Counter_Done              Green_Lamp
```

When C5:0.ACC reaches the preset value of 10, C5:0/DN becomes true. Rung 0001 then energizes Green Lamp O:2/0 to indicate that the target quantity has been reached.

8.3 Rung 0002 - Reset Box Counter

```
|----[ XIC I:1/1 ]-----[ RES C5:0 ]----|
      Reset_Button              Box_Counter
```

When Reset Button I:1/1 becomes true, RES C5:0 resets the counter. The accumulated value returns to zero and the Done status clears, which also turns the Green Lamp OFF.

8.4 Actual RSLogix Implementation

The clean ladder representations above are included for readability. The figure below shows the corresponding ladder logic as implemented in RSLogix Micro Starter Lite.

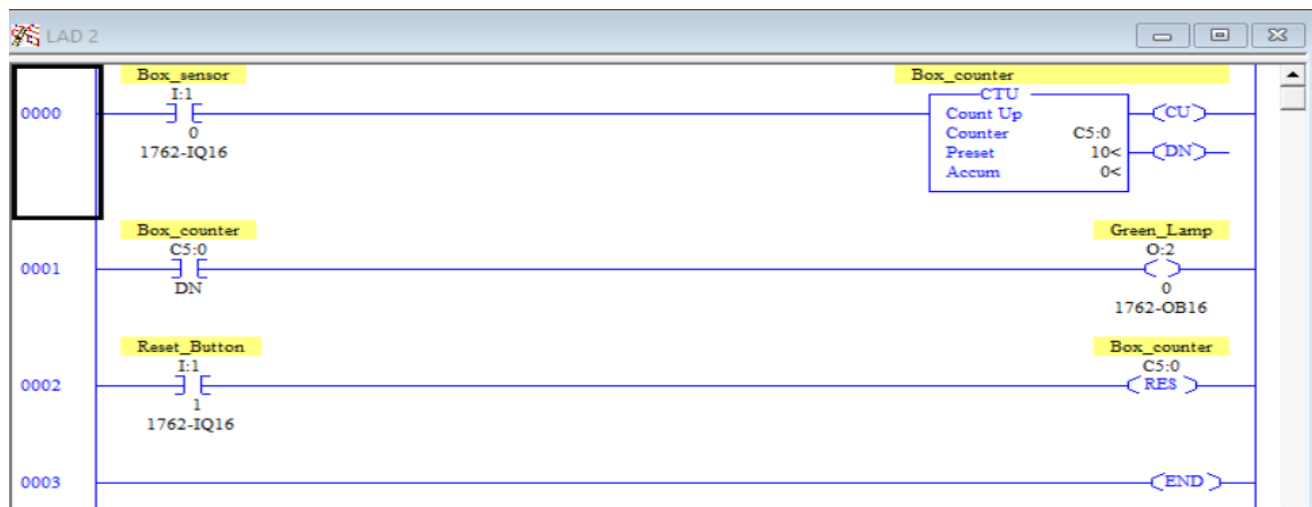


Figure P2.1 - RSLogix Implementation of the Version 1 Automated Box Counting System

9. Sequence of Operation

Step 1 - Initial Condition: C5:0.ACC is zero and Green Lamp O:2/0 is OFF.

Step 2 - Box Detection: A box produces a detection transition at I:1/0.

Step 3 - Counter Increment: C5:0 increments its accumulated value by one.

Step 4 - Continued Counting: Additional valid detection transitions continue increasing the accumulated value.

Step 5 - Target Reached: When C5:0.ACC reaches 10, C5:0/DN becomes true.

Step 6 - Batch Complete Indication: C5:0/DN energizes Green Lamp O:2/0.

Step 7 - Counter Reset: The operator activates Reset Button I:1/1.

Step 8 - New Counting Cycle: RES C5:0 clears the counter, the Done bit returns false, and the Green Lamp turns OFF.

10. Testing & Validation

I tested the completed ladder logic in RSLogix Micro Starter Lite by monitoring the Box Sensor input, counter accumulated value, counter Done bit, Reset input, and Green Lamp output under different operating conditions.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Establish initial condition	C5:0.ACC = 0; C5:0/DN = 0; lamp OFF	Counter zero, Done false, lamp OFF	PASS
T-02	Generate one valid Box Sensor transition	Counter increments 0 to 1	C5:0.ACC increased to 1	PASS
T-03	Continue events while count remains below 10	Counter increments; lamp remains OFF	Counter increased correctly; lamp remained OFF	PASS
T-04	Generate the 10th valid event	ACC = 10; DN = 1; lamp ON	Counter reached 10; Done true; lamp energized	PASS
T-05	Activate Reset Button I:1/1	Counter zero; Done clears; lamp OFF	Counter reset; Done cleared; lamp OFF	PASS
T-06	Generate new event after Reset	New cycle begins from zero	Counter successfully began counting again	PASS

Validation Summary

Testing confirmed that C5:0 correctly accumulated the box-detection events used during the project test. While the accumulated value remained below 10, C5:0/DN remained false and Green Lamp O:2/0 remained OFF. At the target count, the Done bit became true and the lamp energized.

Reset Button I:1/1 successfully executed RES C5:0, returning the accumulated count to zero and clearing the Done status. A new counting cycle then began successfully. All defined Version 1 functional tests passed.

The validation was performed within the project development environment and represents functional PLC program validation, not industrial commissioning or safety validation.

11. Results

The completed Automated Box Counting System successfully performed the required counting, indication, and reset functions. Counter C5:0 accumulated box-detection events from I:1/0, Green Lamp O:2/0 remained OFF below the preset, and the lamp energized when the target quantity of 10 was reached.

Reset Button I:1/1 successfully cleared the accumulated count and Done status, de-energized the Green Lamp, and prepared the system for another counting cycle. All six defined functional tests passed.

12. Challenges & Troubleshooting

The most useful troubleshooting path for this project was: Sensor transition -> CTU accumulator -> Counter Done bit -> Green Lamp.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Counter does not increment	Sensor input does not transition false-to-true	Monitor I:1/0 online	Verify sensor/input state
Counter increments unexpectedly	Sensor signal produces multiple transitions	Monitor I:1/0 while simulating a box	Investigate sensor chatter or signal behavior
Green Lamp turns on too early	Counter preset is incorrect	Check C5:0.PRE	Set preset to 10
Green Lamp never turns on	Counter never reaches preset or Done bit is misaddressed	Monitor C5:0.ACC and C5:0/DN	Verify counter configuration and Rung 0001
Counter does not reset	Reset input or RES instruction is incorrect	Monitor I:1/1 and Rung 0002	Verify Reset input and RES C5:0
Lamp remains ON after Reset	Done bit did not clear or output logic is incorrect	Monitor C5:0/DN and O:2/0	Verify counter reset and output rung

A key lesson was that counter troubleshooting depends on understanding transitions, not simply whether an input is ON or OFF. If the counter does not increment, I need to determine whether a new false-to-true event actually occurred.

13. Lessons Learned

This project helped me understand the difference between time-based and event-based PLC instructions. In Project 1, the result depended on how long a rung remained true. In this project, the important condition was the transition of the counting rung from false to true.

I also learned how the CTU instruction maintains an accumulated value, how the preset establishes a production target, how the Done bit can trigger another function, and how the reset rung deliberately returns stored PLC data to its initial condition.

14. Industrial Relevance

Production counting is widely used in packaging lines, conveyor systems, palletizing operations, warehouse sorting, production monitoring, food and beverage processing, and pharmaceutical packaging.

In a real production system, this counter logic would normally operate as part of a larger machine sequence. Sensor placement, product spacing, signal quality, reset permissions, data retention, production reporting, and downstream machine behavior would also need to be considered.

The Version 1 project therefore represents the foundation of a production-counting function rather than a complete conveyor-control system.

15. Transition to Industrial Version (Version 2)

The Version 2 design can retain the basic counting requirement while expanding it into a more complete production-counting function.

Version 2 Improvement	Engineering Benefit
Adjustable batch target	Allows production quantity to change without modifying ladder logic
HMI current-count display	Provides real-time production progress
Batch-complete indication/alarm	Provides clearer notification at target
Conveyor run interlock at batch completion	Prevents additional production after target where required
Reset permissive	Prevents unintended production-count reset
Sensor diagnostics	Helps identify failed, stuck, or abnormal sensor behavior
Signal validation/debounce logic where required	Reduces false counts caused by poor signal quality or chatter
Batch-complete state	Provides a defined condition for downstream logic
Production statistics	Supports batch totals and operating information
SCADA/MES data integration	Allows production counts to be logged or reported

A CTU already counts false-to-true rung transitions, so a one-shot is not automatically required simply because a sensor remains true across multiple PLC scans. Version 2 should evaluate actual sensor behavior and determine whether edge handling, debounce logic, minimum pulse validation, or other conditioning is justified.

16. Project Summary

The Automated Box Counting System introduced event-based PLC control using a Count Up counter. Box Sensor I:1/0 provided the counting condition, C5:0 stored the current production quantity, and Green Lamp O:2/0 indicated when the preset quantity of 10 had been reached.

A separate Reset input cleared the counter and prepared the system for another counting cycle. The project established my understanding of counter accumulators, presets, Done status, event transitions, reset behavior, and the use of counter status in output logic.

It also created a clear foundation for the Version 2 industrial redesign, where the basic counter can be expanded with operator-adjustable batch quantities, production status, sensor diagnostics, controlled batch completion, and higher-level production-data integration.